INTERNATIONAL SEARCH REPORT

International application No
PCT/US2021/064702

A. CLASSIFICATION OF SUBJECT MATTER
INV. G02B6/30 G02B6/124 G02B6/136 G02B6/42
ADD.

According to International Patent Classification (IPC) or to both national classification and IPC

B. FIELDS SEARCHED

Minimum documentation searched (classification system followed by classification symbols)
G02B

Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched

Electronic data base consulted during the international search (name of data base and, where practicable, search terms used)
EPO-Internal

C. DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
X	US 2020/006304 A1 (CHANG CHIH-CHIEH [TW] ET AL) 2 January 2020 (2020-01-02) figures 2A, 3 paragraphs [0025] - [0028] paragraph [0032] paragraph [0036] -----	1-12
X	US 2020/006088 A1 (YU CHEN-HUA [TW] ET AL) 2 January 2020 (2020-01-02) figure 16 paragraph [0027] paragraph [0038] paragraph [0049] paragraph [0050] -----	1-12

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[X] Further documents are listed in the continuation of Box C.

[X] See patent family annex.

* Special categories of cited documents :

"A" document defining the general state of the art which is not considered to be of particular relevance

"E" earlier application or patent but published on or after the international filing date

"L" document which may throw doubts on priority claim(s) or which is cited to establish the publication date of another citation or other special reason (as specified)

"O" document referring to an oral disclosure, use, exhibition or other means

"P" document published prior to the international filing date but later than the priority date claimed

"T" later document published after the international filing date or priority date and not in conflict with the application but cited to understand the principle or theory underlying the invention

"X" document of particular relevance;; the claimed invention cannot be considered novel or cannot be considered to involve an inventive step when the document is taken alone

"Y" document of particular relevance;; the claimed invention cannot be considered to involve an inventive step when the document is combined with one or more other such documents, such combination being obvious to a person skilled in the art

"&" document member of the same patent family

Date of the actual completion of the international search	Date of mailing of the international search report
17 March 2022	19/05/2022

Name and mailing address of the ISA/	Authorized officer
European Patent Office, P.B. 5818 Patentlaan 2 NL - 2280 HV Rijswijk Tel. (+31-70) 340-2040, Fax: (+31-70) 340-3016	Dregely, Daniel

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Form PCT/ISA/210 (second sheet) (April 2005)

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Box No. II Observations where certain claims were found unsearchable (Continuation of item 2 of first sheet)

This international search report has not been established in respect of certain claims under Article 17(2)(a) for the following reasons:

1. ☐ Claims Nos.:
because they relate to subject matter not required to be searched by this Authority, namely:

2. ☐ Claims Nos.:
because they relate to parts of the international application that do not comply with the prescribed requirements to such an extent that no meaningful international search can be carried out, specifically:

3. ☐ Claims Nos.:
because they are dependent claims and are not drafted in accordance with the second and third sentences of Rule 6.4(a).

Box No. III Observations where unity of invention is lacking (Continuation of item 3 of first sheet)

This International Searching Authority found multiple inventions in this international application, as follows:

see additional sheet

1. ☐ As all required additional search fees were timely paid by the applicant, this international search report covers all searchable claims.

2. ☐ As all searchable claims could be searched without effort justifying an additional fees, this Authority did not invite payment of additional fees.

3. ☐ As only some of the required additional search fees were timely paid by the applicant, this international search report covers only those claims for which fees were paid, specifically claims Nos.:

4. ☒ No required additional search fees were timely paid by the applicant. Consequently, this international search report is restricted to the invention first mentioned in the claims;; it is covered by claims Nos.:

1-12

Remark on Protest

☐ The additional search fees were accompanied by the applicant's protest and, where applicable, the payment of a protest fee.

☐ The additional search fees were accompanied by the applicant's protest but the applicable protest fee was not paid within the time limit specified in the invitation.

☐ No protest accompanied the payment of additional search fees.

FURTHER INFORMATION CONTINUED FROM PCT/ISA/ 210

This International Searching Authority found multiple (groups of) inventions in this international application, as follows:

1. claims: 1-12

A semiconductor chip package comprising a groove for aligning an optical fiber and corresponding fabrication method

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2. claims: 13-22

A semiconductor chip package comprising a lens and grating for coupling an optical fiber and corresponding fabrication method

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C(Continuation). DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
A	CN 112 034 567 A (NAT CT ADVANCED PACKAGING CO LTD NCAP CHINA) 4 December 2020 (2020-12-04) figures 5, 11	1-12
A	US 2017/254968 A1 (DING LIANG [SG] ET AL) 7 September 2017 (2017-09-07) figures 1, 5D	1-12

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INTERNATIONAL SEARCH REPORT

Information on patent family members

International application No
PCT/US2021/064702

Patent document cited in search report	Publication date	Patent family member(s)	Publication date
US 2020006304 A1	02-01-2020	CN 110646898 A	03-01-2020
		DE 102019116579 A1	02-01-2020
		KR 20200001536 A	06-01-2020
		TW 202014746 A	16-04-2020
		US 2020006304 A1	02-01-2020
		US 2021202453 A1	01-07-2021
US 2020006088 A1	02-01-2020	CN 110648927 A	03-01-2020
		DE 102019105763 A1	02-01-2020
		KR 20200001473 A	06-01-2020
		TW 202002052 A	01-01-2020
		US 2020006088 A1	02-01-2020
		US 2020411333 A1	31-12-2020
CN 112034567 A	04-12-2020	CN 112034567 A	04-12-2020
		WO 2022048124 A1	10-03-2022
US 2017254968 A1	07-09-2017	CN 206546453 U	10-10-2017
		US 2017254968 A1	07-09-2017
		US 2018275359 A1	27-09-2018